

CloudFront Performance Optimization using Amazon S3 Static Website Hosting

To Begin with the Lab

Summary of the Lab

In this lab, a static website was hosted using **Amazon S3 Static Website Hosting** and then accelerated globally using **Amazon CloudFront**. The website consisted of an HTML page, image assets, and a video file stored inside an S3 bucket hosted in the N. Virginia region. Performance was first measured using the S3 website endpoint directly to observe latency from different geographic locations. A CloudFront distribution was then created using the S3 website endpoint as the origin, allowing content to be cached across AWS edge locations worldwide. Finally, website performance was tested again to verify reduced latency and improved global user experience through content delivery network caching.

Understanding Amazon CloudFront

Amazon CloudFront is a Content Delivery Network (CDN) that caches content at AWS edge locations around the world so users receive data from the nearest location instead of the original server. Without CloudFront, every request must travel directly to the origin server, increasing latency for users located far away from the hosting region. CloudFront reduces response time by storing copies of frequently accessed content closer to end users while still retrieving updates from the origin when necessary. This improves website speed, scalability, and availability globally.

Example:

A website is hosted in an **Amazon S3** bucket located in **N. Virginia**. Users in Virginia access the site with almost no delay, but users in Australia experience nearly 200 ms latency. After enabling **Amazon CloudFront**, the same website content is served from nearby edge locations, reducing latency to only a few milliseconds regardless of user location.

Understanding Amazon S3 Static Website Hosting

Amazon S3 Static Website Hosting allows HTML, CSS, JavaScript, images, and videos to be served directly from an S3 bucket without requiring a web server. The service automatically provides a website endpoint that can host public-facing static websites. While this approach is simple and cost-effective, all requests must still reach the AWS Region where the bucket resides.

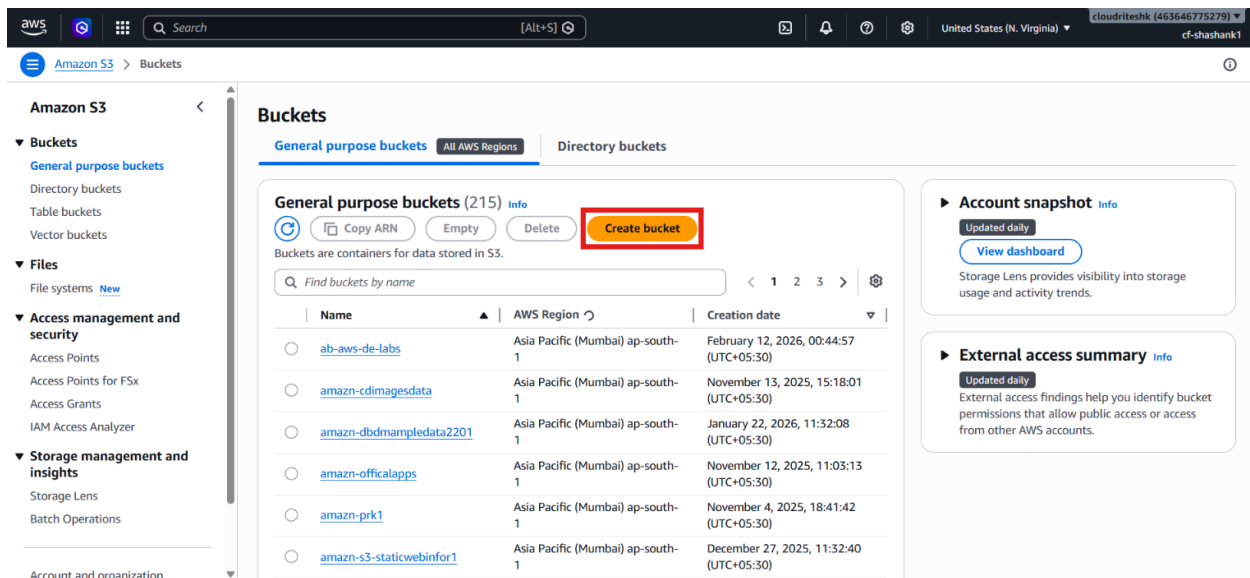
Example:

A bucket named **cloudfox-web** stores **index.html**, **intro.mp4**, and **wallpaper.png**. Users can access the website directly through the S3 website endpoint, but visitors far from N. Virginia experience higher latency because every request must travel to that single AWS Region.

Lab Steps

1. Create an S3 Bucket for Website Hosting

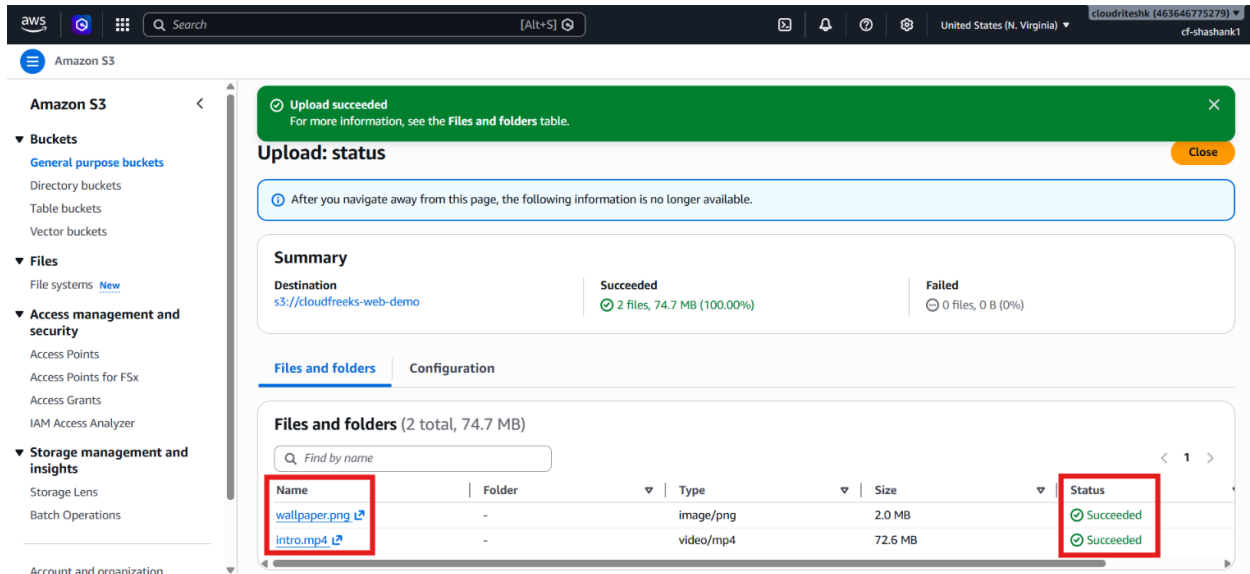
- Sign in to the AWS Management Console.
- Navigate to **Amazon S3**.
- Switch to the **N. Virginia (us-east-1)** region.
- Click **Create bucket**.



- Configure the bucket: Bucket Name: **cloudfreeks-web-demo**,
Region: **us-east-1**
- Leave all remaining settings as default.
- Click **Create bucket**.

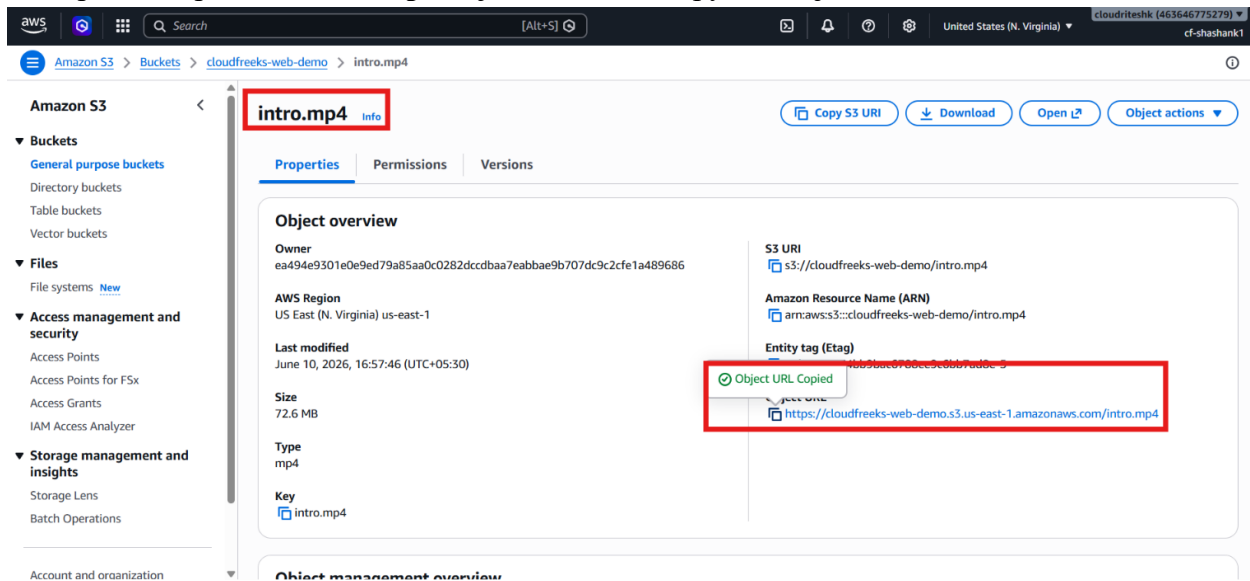
Upload Website Content

- Prepare website files locally: **index.html**, **intro.mp4**, **wallpaper.png**
- Open the created S3 bucket.
- Click **Upload** → **Add files**.
- Upload: **intro.mp4**, **wallpaper.png**
- Wait for both uploads to complete.

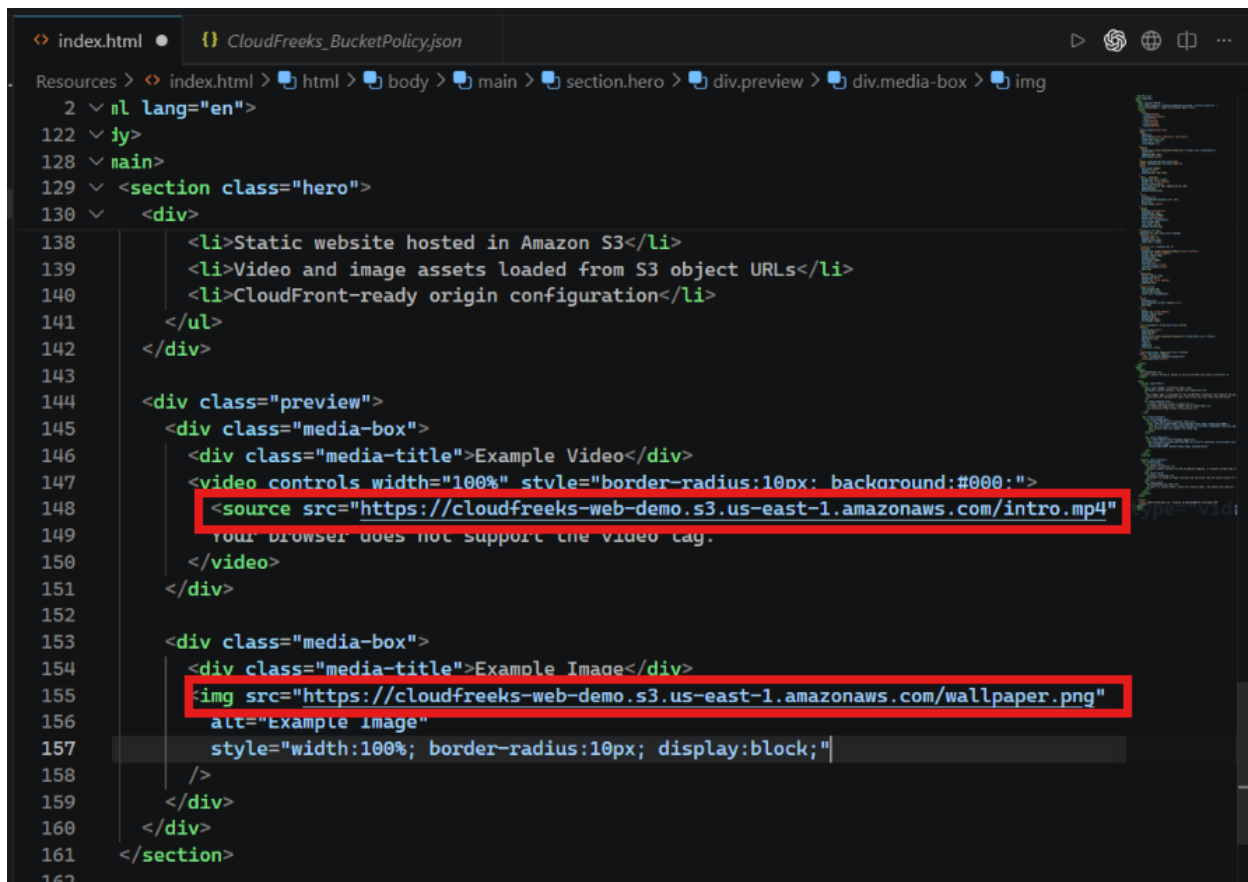


Update Website Asset URLs

- Open **index.html** using a code editor.
- Open the uploaded **intro.mp4** object in S3 and copy its **Object URL**.



- Replace the video path inside **index.html** with the S3 Object URL.
- Do the same for **wallpaper.png**



```
index.html • CloudFreeeks_BucketPolicy.json
Resources > index.html > html > body > main > section.hero > div.preview > div.media-box > img
  2 <html lang="en">
122 </html>
128 <main>
129 <section class="hero">
130 <div>
138 <li>Static website hosted in Amazon S3</li>
139 <li>Video and image assets loaded from S3 object URLs</li>
140 <li>CloudFront-ready origin configuration</li>
141 </ul>
142 </div>
143
144 <div class="preview">
145 <div class="media-box">
146 <div class="media-title">Example Video</div>
147 <video controls width="100%" style="border-radius:10px; background:#000:">
148 <source src="https://cloudfreeks-web-demo.s3.us-east-1.amazonaws.com/intro.mp4" type="video/mp4">
149 Your browser does not support the video tag.
150 </video>
151 </div>
152
153 <div class="media-box">
154 <div class="media-title">Example Image</div>
155 
156
157 </div>
158
159 </div>
160 </div>
161 </section>
162
```

- Save the **index.html** file.

Upload Updated HTML File

- Return to the S3 bucket.
- Click **Upload** → **Add files**.
- Upload: **index.html**
- Verify all website files exist in the bucket.

Enable Static Website Hosting

- Navigate to **Amazon S3** → open the bucket.
- Select the **Properties** tab.

Amazon S3 > Buckets > cloudfreeks-web-demo

cloudfreeks-web-demo Info

Objects | Metadata | **Properties** | Permissions | Metrics | Management | File systems - new | Access Points

Bucket overview

AWS Region: US East (N. Virginia) us-east-1 | Amazon Resource Name (ARN): arn:aws:s3:::cloudfreeks-web-demo | Creation date: June 10, 2026, 16:52:46 (UTC+05:30)

Bucket Versioning [Edit](#)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Disabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

Bucket ABAC [Edit](#)

Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and

- **Scroll to Static website hosting.**

Amazon S3 > Buckets > cloudfreeks-web-demo

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

Object Lock
Disabled

Requester pays [Edit](#)

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Static website hosting [Edit](#)

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#) [Create Amplify app](#)

S3 static website hosting
Disabled

- **Click Edit → Select Enable.**
- **Configure: Index document: index.html**

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
 Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
 Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

- Click **Save changes**.
- Copy the generated **Static Website Endpoint URL (Bucket website endpoint)**.

Successfully edited static website hosting.

Requester pays
Disabled

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using **AWS Amplify Hosting** for static website hosting. Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#). [Create Amplify app](#)

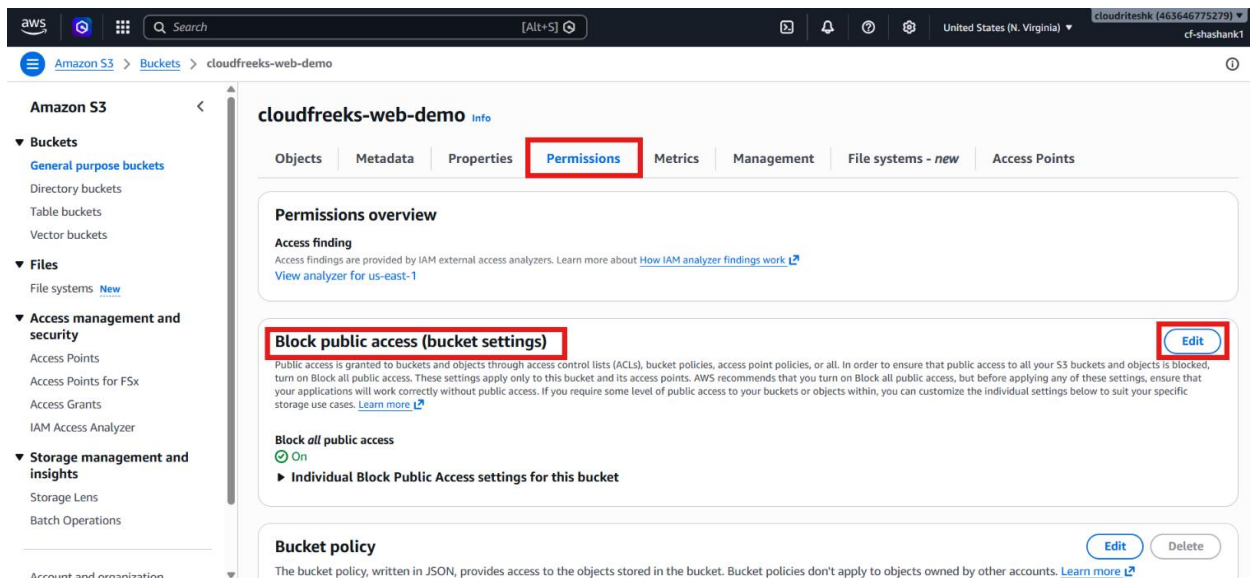
S3 static website hosting
Enabled

Hosting type
Bucket hosting

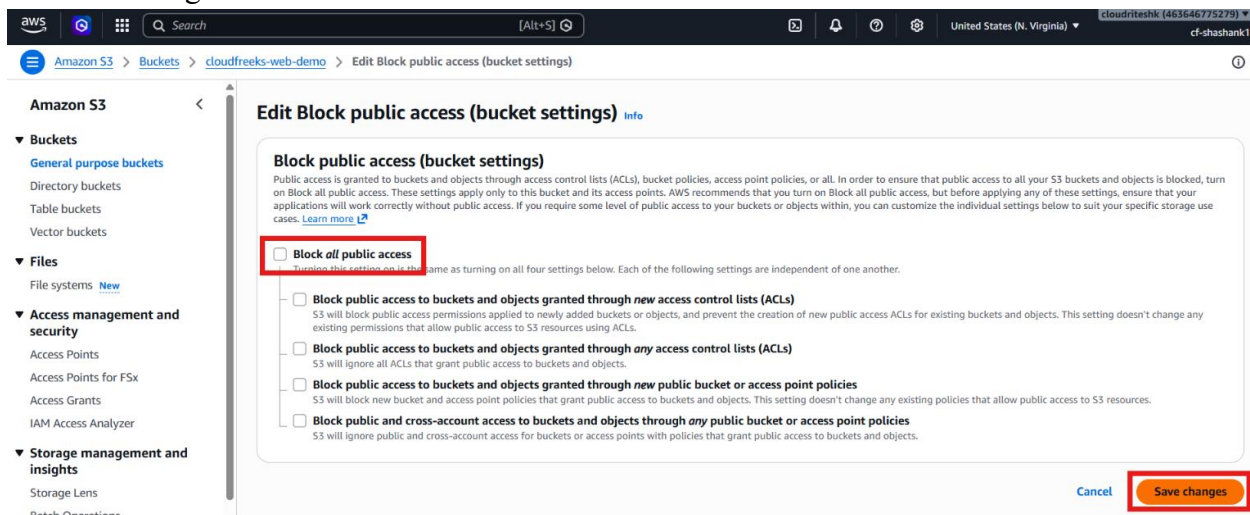
Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS region-specific website endpoint of the bucket. [Learn more](#)
<http://cloudfreeks-web-demo.s3-website-us-east-1.amazonaws.com>

Make the Bucket Public

- Open the **Permissions** tab.
- Locate **Block Public Access** → Click **Edit**.

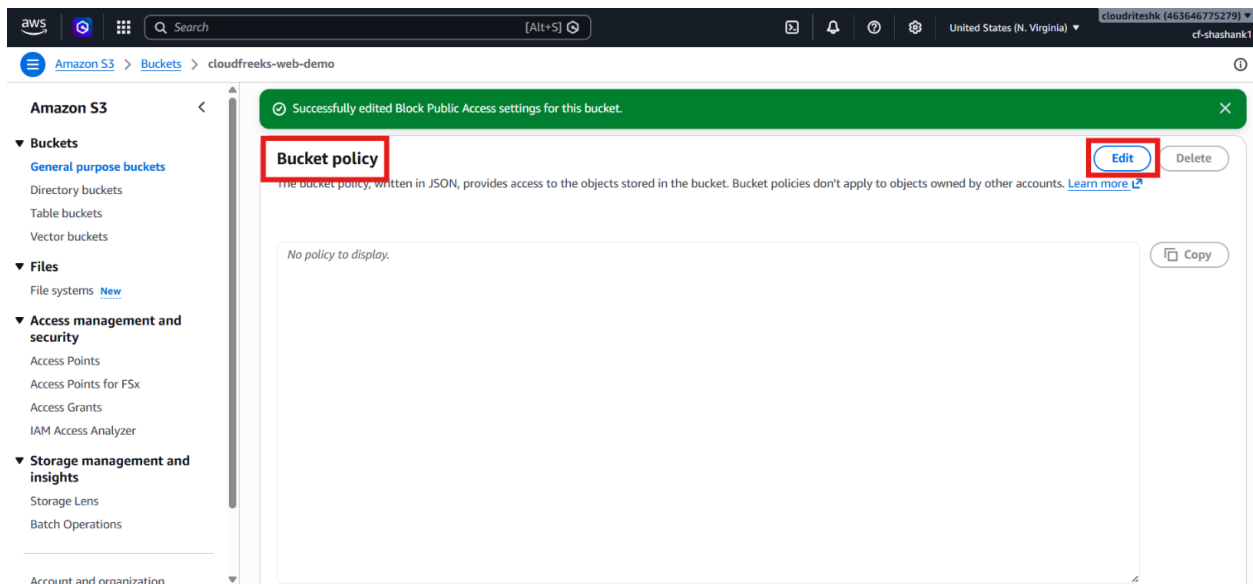


- Disable **Block all public access**.
- Confirm the warning message.
- Save changes.



Configure Bucket Policy

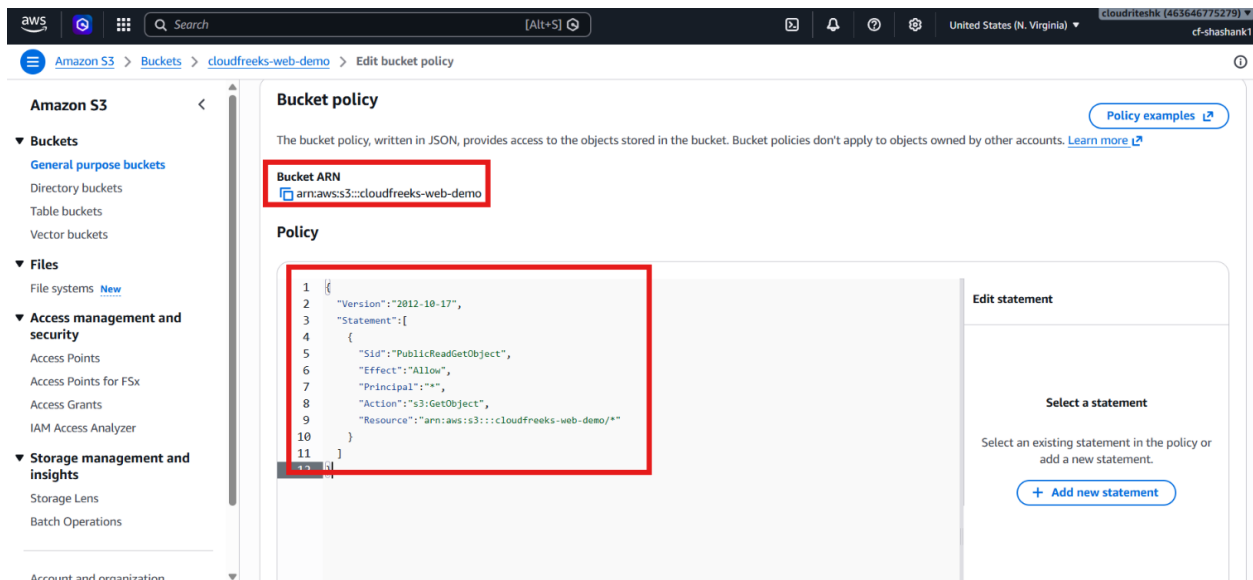
- Open **Permissions** → **Bucket Policy**.
- Click **Edit**.



- Add a public read bucket policy.
- Replace the bucket ARN with the ARN of your bucket.

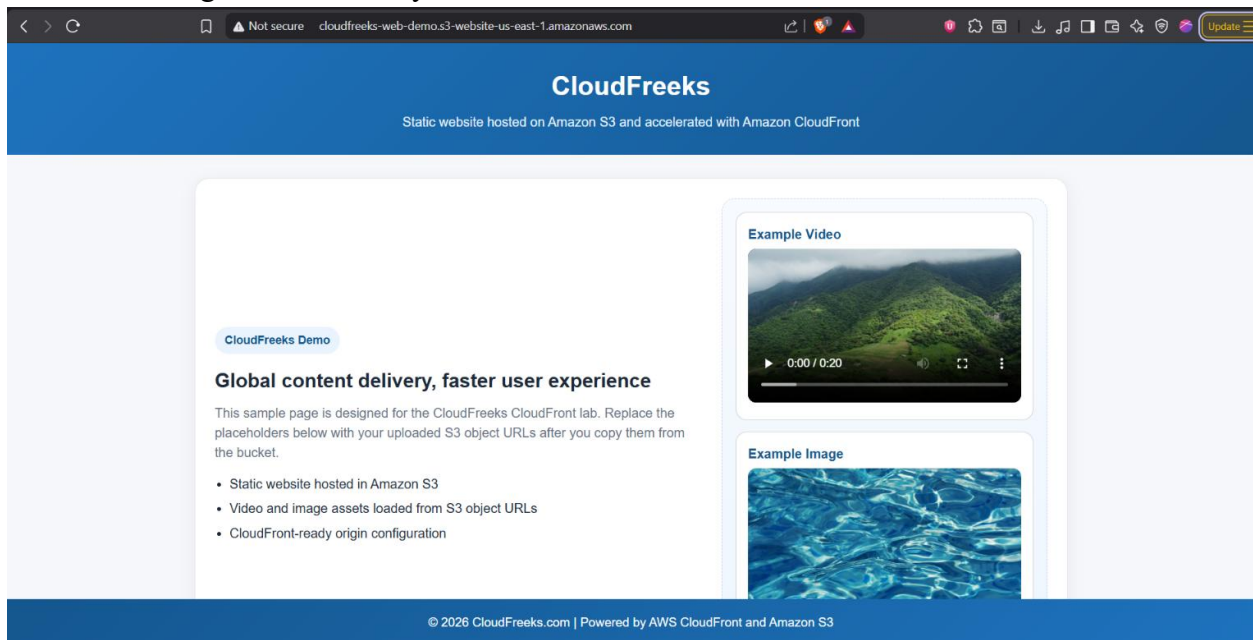
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::cloudfreeks-web-demo/*"
    }
  ]
}
```

- Save the bucket policy.



Verify S3 Website Hosting

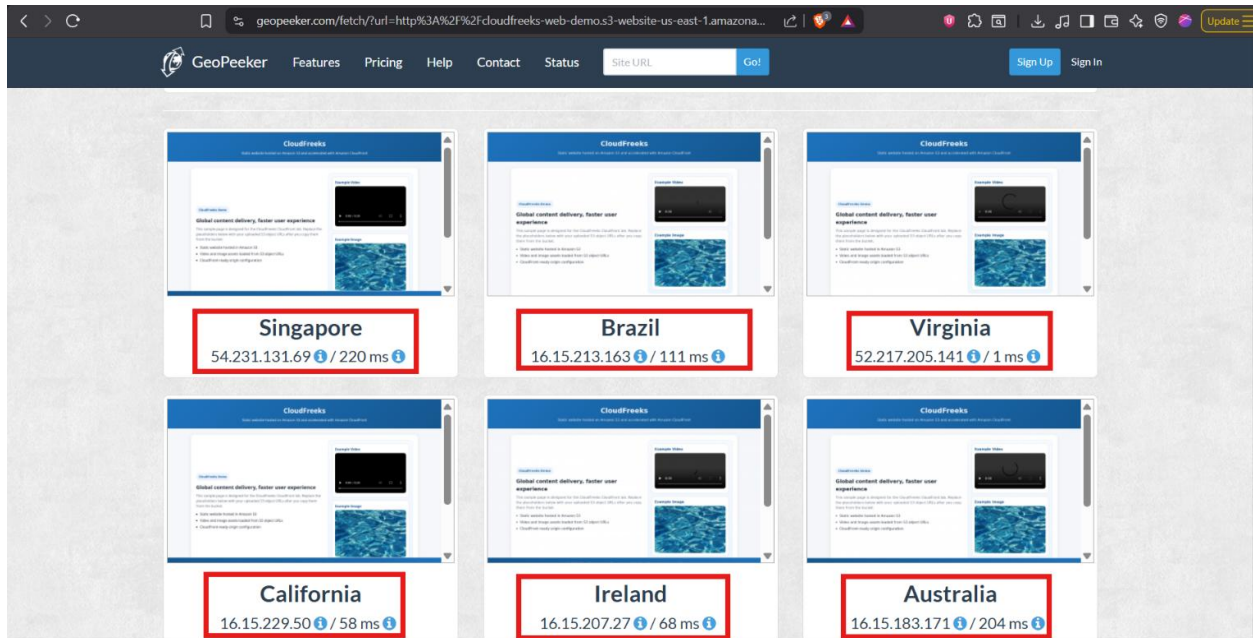
- Navigate to **Properties**.
- Copy the **Static Website Endpoint URL**.
- Open the URL in a browser.
- Verify: Website loads successfully,
Video plays correctly,
Images load correctly



Test Website Performance Without CloudFront

- Open a website performance testing tool like [geopeeker.com](https://www.geopeeker.com)
- Paste the S3 Website Endpoint URL.

- Run tests from multiple geographic locations.
- Observe latency differences between regions.



- Record the results.

Example observations:

Location	Latency
Virginia	~ 1 ms
California	~ 58 ms
Ireland	~ 68 ms
Brazil	~ 111 ms
Australia	~ 204 ms

2. Create a CloudFront Distribution

Understanding Origin Configuration

When creating a CloudFront distribution, the origin identifies where CloudFront retrieves the original content. In this lab, the S3 Static Website Endpoint acts as the origin. CloudFront caches copies of that content at AWS edge locations worldwide and serves users from the nearest location whenever possible.

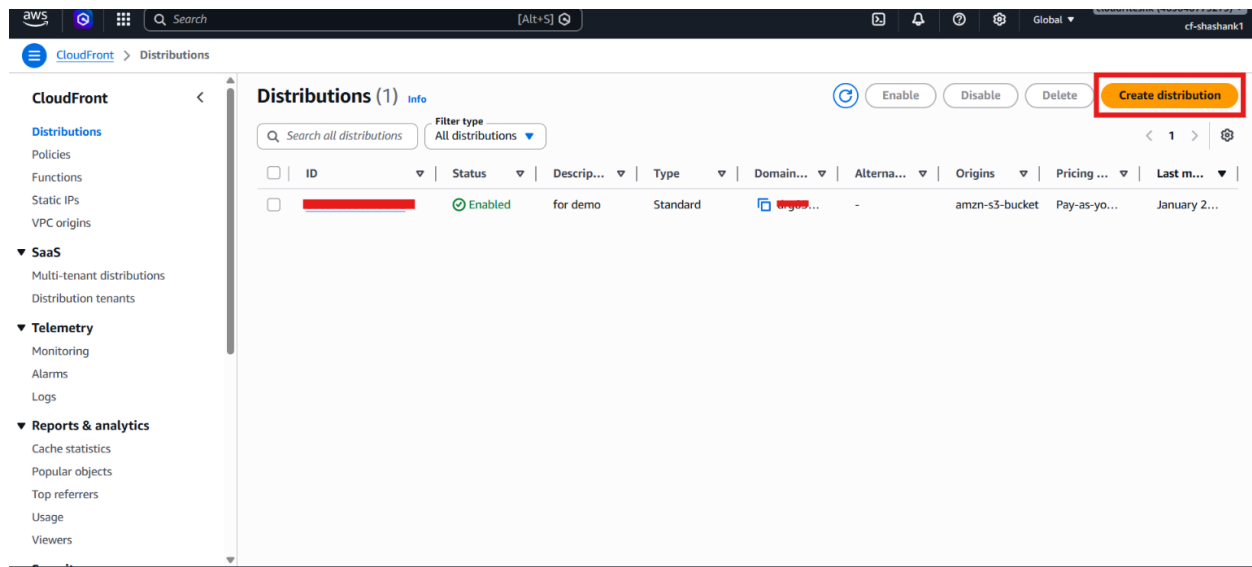
Example:

The website files remain stored in the **cloudfox-web** bucket. When a user in Australia accesses the site, CloudFront retrieves the content from S3 once and caches it at a nearby edge location. Future requests are served directly from the edge cache instead of N. Virginia.

Create the Distribution

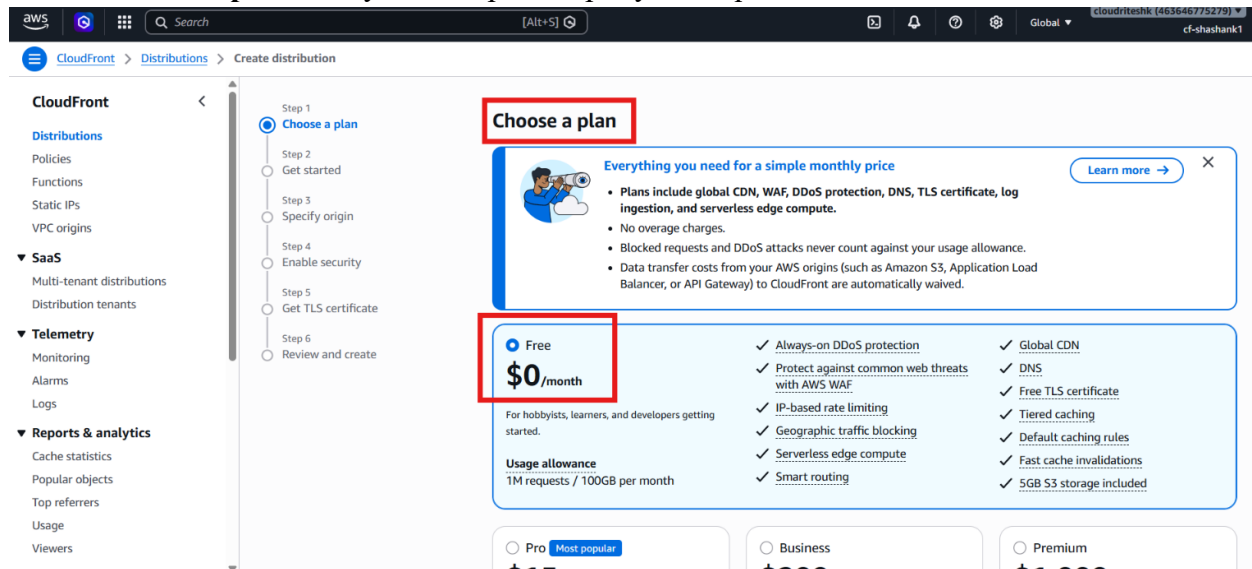
- Navigate to **Amazon CloudFront**.

- Click **Create distribution**.



Configure Plan

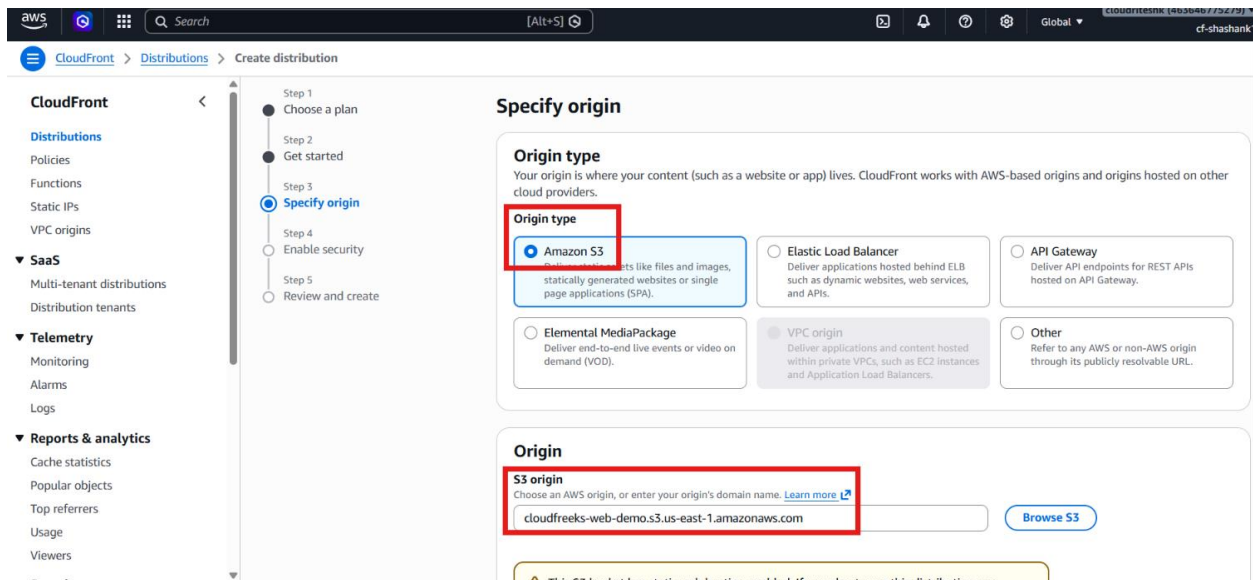
- Choose **Free plan** or any desired plan as per your request and click **Next**.



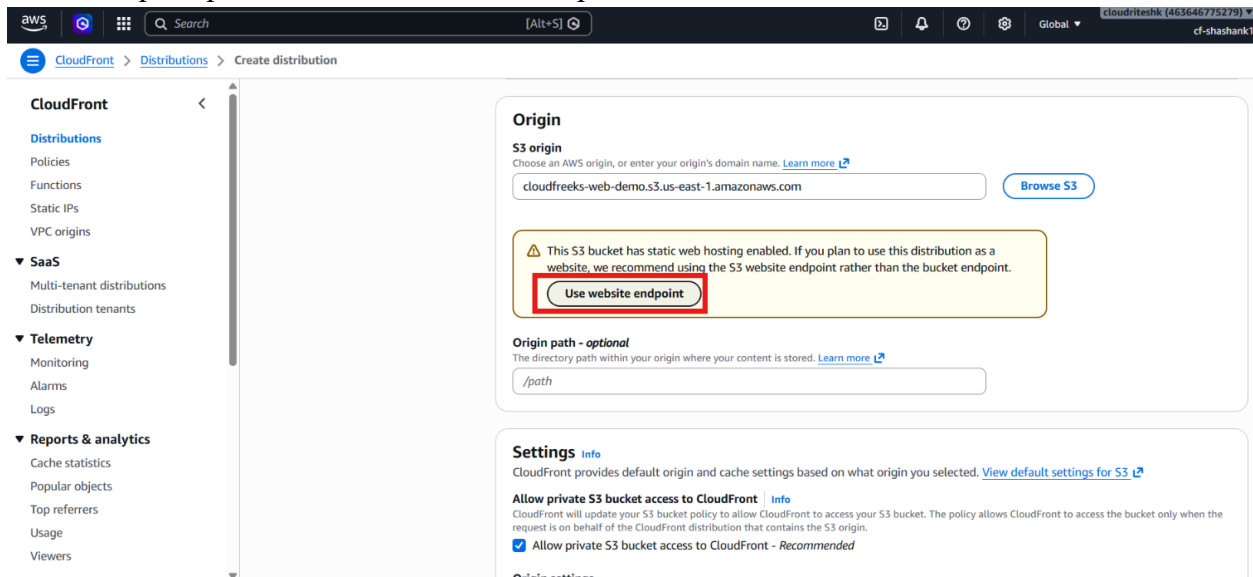
- Enter **Distribution name** and click **Next**.

Configure Origin

- Under **Origin type**, select **Amazon S3**.
- Under **Origin**, select the S3 bucket.



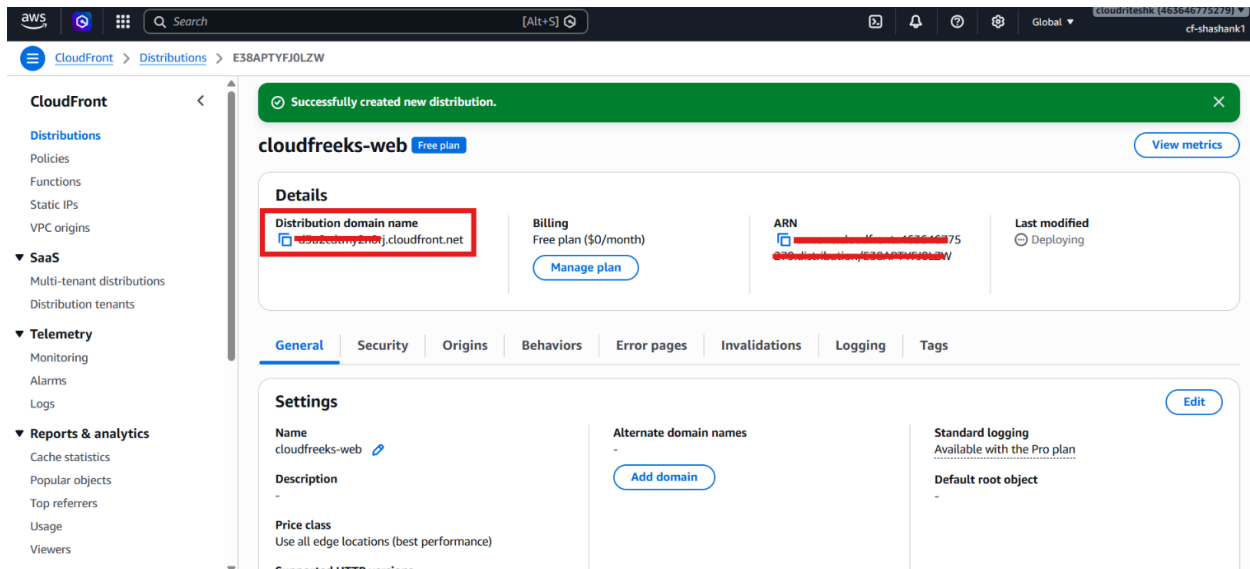
- When prompted, select: Use website endpoint.



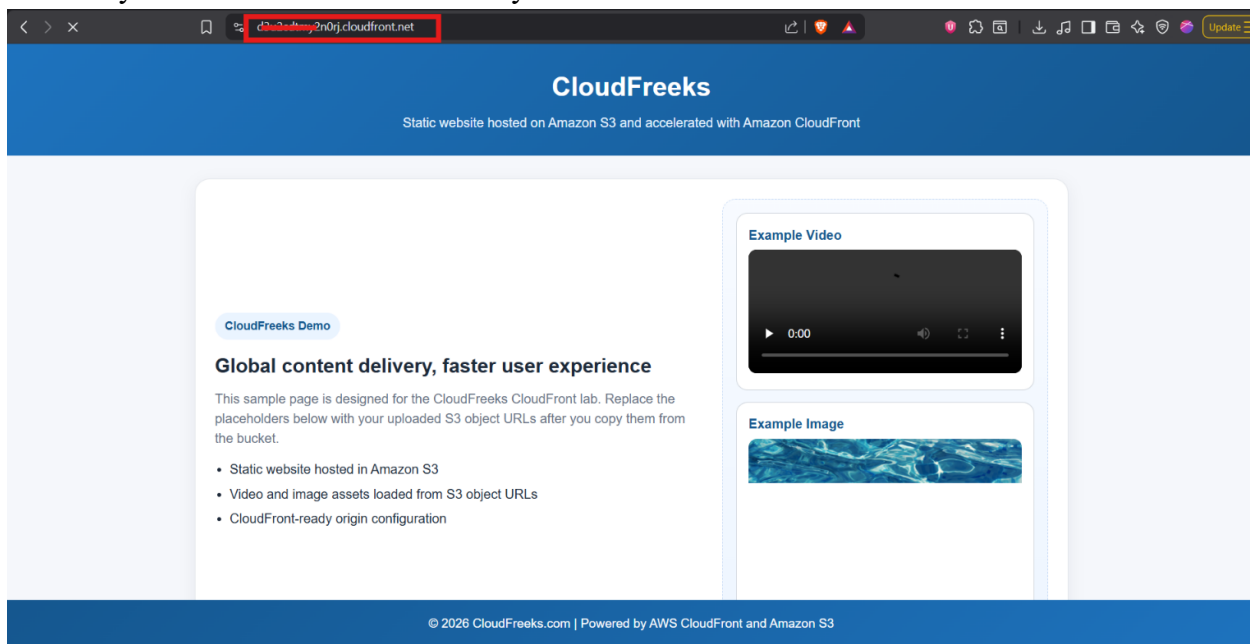
- Leave **Origin Path** empty because the website files are stored at the root of the bucket.
- Leave rest settings as default.
- Click **Create Distribution**.
- Wait for deployment to complete.
- Verify the distribution status becomes: Enabled

Obtain CloudFront URL

- Open the newly created distribution.
- Copy the generated CloudFront domain name.
Example: dxxxxxxxxxxxxx.cloudfront.net

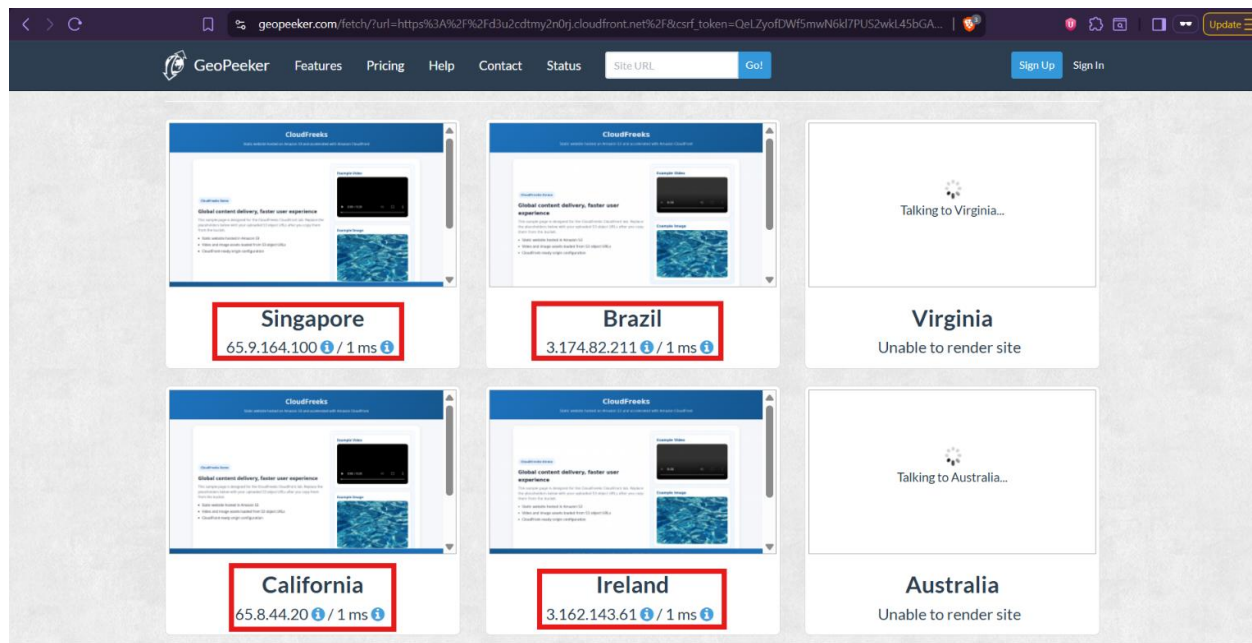


- Open the CloudFront URL in a browser.
- Verify the website loads successfully.



Test Website Performance with CloudFront

- Open the same performance testing tool.
- Paste the CloudFront URL.



- Run tests from multiple global locations.
- Compare the results with the S3 endpoint.

Example observations:

Location	Latency
Singapore	1 ms
California	1 ms
Ireland	1 ms
Brazil	1 ms

- Verify global latency is significantly reduced.

Verify CloudFront Benefits

- Confirm the website loads through the CloudFront URL.
- Confirm users receive content from nearby edge locations.
- Verify reduced latency from geographically distant locations.
- Confirm improved global performance compared to direct S3 access.

Verification

- The website is hosted successfully using **Amazon S3 Static Website Hosting**.
- Static content is publicly accessible through the S3 website endpoint.
- **Amazon CloudFront** uses the S3 website endpoint as its origin.
- Content is cached across AWS edge locations.
- Website latency is reduced globally.
- Users receive faster content delivery regardless of geographic location.